ICT Academy of Kerala · Python Programming

QuickMart Billing System

Python Functions · 1-hour case study

Submit your work on Paatshala as a GitHub repository link

How to work

Step	What to do
1	Create a project folder in VS Code and activate your virtual environment. The terminal must show the environment name before you run anything.
2	Create one file per case study, using the file name given at the top of each case study.
3	Copy the starter data exactly as given. Do the tasks in order and run your program after every task.
4	Your output must match the expected output exactly, including spaces and alignment.
5	Call your trainer when you finish, and try the challenge if time is left.

The scenario

QuickMart is a campus store. The owner wants a billing program built from small, reusable **functions** so that prices, discounts and tax can change without rewriting the program. **You will build it.**
File name: `quickmart_billing.py` **Time:** 60 minutes

Skills you will use

- Defining functions with parameters, default values and return values
- Functions that process a list of dictionaries
- Passing a function to `sorted` with `key=`
- Formatted output and testing with `assert`

STARTER DATA (COPY EXACTLY)

```
# quickmart_billing.py - QuickMart Billing System
cart = [
    {"name": "Notebook", "category": "Stationery", "price": 40,  "qty": 5},
    {"name": "Pen",      "category": "Stationery", "price": 10,  "qty": 11},
    {"name": "Biscuits", "category": "Snacks",     "price": 25,  "qty": 4},
    {"name": "Juice",    "category": "Drinks",     "price": 45,  "qty": 3},
    {"name": "Chips",    "category": "Snacks",     "price": 20,  "qty": 6},
]
```

Functions you will write

#	Function	What it does
1	line_total(item)	returns price × qty of one item
2	subtotal(cart)	returns the sum of all line totals (use `line_total`)
3	apply_discount(amount, percent=0)	returns the amount after reducing it by percent; default 0
4	gst(amount, rate=18)	returns the GST on the amount; default rate 18
5	category_totals(cart)	returns a dictionary of category to total value
6	top_items(cart, n=2)	returns the names of the n items with the highest line total (use `sorted` with `key=line_total`)
7	print_bill(cart, discount_percent=0)	prints the bill table and the summary, using the functions above
8	main()	calls everything and prints each task heading; then ends with a few `assert` tests

Tasks and expected output

Task 1 · Line total

Print the line total of the first item in the cart.

EXPECTED OUTPUT

```
Task 1: Line total
200
```

Task 2 · Subtotal

Print the subtotal of the cart.

EXPECTED OUTPUT

```
Task 2: Subtotal
665
```

Task 3 · Discount

Print `apply_discount(500)` and `apply_discount(500, 10)`.

EXPECTED OUTPUT

```
Task 3: Discount
500.0
450.0
```

Task 4 · GST

Print `gst(100)` and `gst(100, 5)`.

EXPECTED OUTPUT

```
Task 4: GST
18.0
5.0
```

Task 5 · Category totals

Print the category totals, categories in alphabetical order, the name padded to 12 characters.

EXPECTED OUTPUT

```
Task 5: Category totals
Drinks      135
Snacks      220
Stationery  310
```

Task 6 · Top items

Print `top_items(cart)`, then `top_items(cart, 3)`.

EXPECTED OUTPUT

```
Task 6: Top items
['Notebook', 'Juice']
['Notebook', 'Juice', 'Chips']
```

Task 7 · Bill

Call `print_bill(cart, 10)`. Columns: item 10 wide (left), price 6, qty 5, total 8 (right aligned); the summary amounts are right aligned in 8 characters with 2 decimals. The bill has a 10% discount and 18% GST on the discounted amount.

EXPECTED OUTPUT

```
Task 7: Bill
Item       Price  Qty   Total
----------------------------
Notebook      40    5     200
Pen           10   11     110
Biscuits      25    4     100
Juice         45    3     135
Chips         20    6     120
----------------------------
Subtotal                665.00
Discount 10%             66.50
GST 18%                 107.73
Grand total             706.23
```

Task 8 · Tests

Write four `assert` tests (for example line_total of one item, apply_discount(200, 50), gst(200), and the top item) and print the message below if they all pass.

EXPECTED OUTPUT

```
Task 8: Tests
All tests passed
```

Print the heading lines (for example `Task 1: Line total`) exactly as shown, with a blank line between tasks. Write and run each function before moving to the next.

Suggested time

Tasks	Minutes
Set up folder, virtual environment, copy data	5
Functions 1 to 4	15
Functions 5 and 6	15
Function 7 (the bill) and main()	20
Check output, push to GitHub, submit link on Paatshala	5

Challenge (optional)

- Add a `highest_category(cart)` function that returns the category with the highest total.
- Add a `coupon` parameter to `print_bill` that takes a further flat amount off.
- Print the bill for two different discounts using the same function.

Submission

- Create a GitHub repository and upload `quickmart_billing.py`.
- Submit the repository link on Paatshala.